Joint Topology Identification and Signal Inference by SEM

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***Abstract*—Network topology identification and signal inference are fundamental problems in graph signal processing. While Structural Equation Models (SEMs) effectively capture causal dependencies in directed networks, most existing methods assume known exogenous inputs. To address this limitation, we propose a generalized SEM that incorporates sparse nodal and temporal excitation priors, enabling joint recovery of network topology and latent signals. An efficient Alternating Direction Method of Multipliers (ADMM)-based solver is developed to tackle the resulting nonconvex problem. Experiments on synthetic and real data demonstrate clear improvements in accuracy and robustness over conventional SEM approaches.**

***Index Terms*—Alternating direction method of multipliers (ADMM), directed network topology identification, structural equation model (SEM).**

I. Introduction

Graph signal processing (GSP) [1]–[6] provides a principled framework for analyzing data on non-Euclidean domains by modeling signals on graph nodes and their dependencies via edges, underscoring the importance of inferring graph structures from observed signals in large-scale complex networks. SEM [7]–[9] has emerged as an effective approach for directed network analysis, as it explicitly models data generation to identify causal dependencies, and has been applied in social, genomic, and neural network inference [10]–[15].

From a broader signal processing perspective, array signals—traditionally analyzed using explicit physical models such as array geometry and wave propagation—can also be viewed as multichannel signals defined over graph nodes. Unlike classical array processing, SEM adopts a data-driven modeling paradigm that captures inter-sensor dependencies without relying on explicit physical priors, making it well suited for scenarios where such models are unknown.

Most conventional SEM employ maximum likelihood or least-squares (LS) optimization for network inference [7], [8], [11], [12], [16], [17]. While LS is computationally efficient [11], [12], [16], [17], it assumes error-free inputs and account only for output noise, an assumption rarely met in practice. This assumption is seldom satisfied in real-world, where sensors and measurement protocols introduce uncertainties into both input and output variables. To address this, more robust estimators like the total least squares (TLS) method [18]–[20] have been proposed. By accounting for noise in both exogenous and endogenous variables, TLS-based SEM yields more reliable topology estimates in noisy environments [21].

Nevertheless, existing TLS-based SEM methods generally assume fully observed exogenous inputs, which is unrealistic in many applications where external excitations are partially observed or latent, leading to biased topology estimation and degraded signal recovery. To overcome this limitation, we propose a generalized SEM that jointly infers the directed network topology and unknown exogenous inputs by exploiting sparsity in the nodal domain and structured temporal excitation patterns. An efficient ADMM-based algorithm is developed to solve the resulting problem, and experiments on synthetic and real-world datasets demonstrate the effectiveness and practical utility of the proposed approach.

The remainder of this paper is organized as follows. Section II presents the proposed SEM framework and formulates the joint inference problem along with its solution. Section III reports the experimental results and analysis, and Section IV concludes the paper.

II. Joint Topology Identification and Exogenous Input Inference

A. Network Model and Problem Formulation

Consider a directed network of N nodes represented by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$. The entry $A_{i,j} \neq 0$ denotes a directed edge from node j to node i, and $A_{i,i} = 0$. Let $y_i(t)$ represent the observation at node i at time t. Under the SEM framework, node dynamics follow

$$y_i(t) = \sum_{j\neq i} A_{i,j} y_j(t) + x_i(t), \quad t = 1, \ldots, T, \tag{1}$$

where $x_i(t)$ aggregates all exogenous inputs acting on node i. Defining $\mathbf{y}(t) = [y_1(t), \ldots, y_N(t)]^\top$ and $\mathbf{x}(t) = [x_1(t), \ldots, x_N(t)]^\top$, the model compactly reads $\mathbf{y}(t) = \mathbf{A}\mathbf{y}(t) + \mathbf{x}(t)$. Collecting all T time steps, the model becomes $\mathbf{Y} = \mathbf{A}\mathbf{Y} + \mathbf{X}$ where $\mathbf{X} = [\mathbf{x}(1), \ldots, \mathbf{x}(T)]$ and $\mathbf{Y} = [\mathbf{y}(1), \ldots, \mathbf{y}(T)]$.

To model realistic excitation patterns, we assume that (i) each exogenous source may affect multiple nodes and each node may be influenced by multiple sources, and (ii) inputs

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exhibit distinct onset times and durations. Thus, the SEM-based signal model is expressed as

$$\mathbf{Y} = \mathbf{A}\mathbf{Y} + \sum_{q=1}^{Q} \mathbf{X}_q = \mathbf{A}\mathbf{Y} + \sum_{q=1}^{Q} \mathbf{d}_q \mathbf{c}_q^\top, \tag{2}$$

where $Q \ll N$ represents the number of latent sources. The vector $\mathbf{d}_q \in \mathbb{R}^N$ captures the spatial distribution of the q-th input (with nonzero entries indicating excited nodes), while $\mathbf{c}_q \in \mathbb{R}^T$ represents its temporal excitation pattern.

In many real-world scenarios, exogenous inputs persist over continuous time intervals, resulting in a block-sparse (piecewise constant) structure in $\mathbf{c}_q$. To exploit this, we employ the first-order temporal difference operator $\mathbf{D}_T \in \mathbb{R}^{T\times(T-1)}$, defined as a bidiagonal matrix with -1 on the main diagonal and 1 on the first super-diagonal. The sparsity of the transformed vector $\mathbf{D}_T^\top \mathbf{c}_q$ effectively captures the transitions (onsets and offsets) of the exogenous signals. This differencing operation can be extended row-wise to the aggregate input matrix $\mathbf{X}$ to capture temporal dynamics across the network.

Since $\mathbf{X}$ is decomposed into Q rank-one matrices, its rank is at most Q. As directly enforcing a rank constraint is computationally intractable, we adopt nuclear-norm regularization ($\|\mathbf{X}\|_*$) as a convex proxy. Furthermore, because each $\mathbf{d}_q$ is sparse—implying that only a subset of nodes is excited by any single source—the matrix $\mathbf{X}$ exhibits a row-sparse structure. We promote this via an $\ell_{1,2}$ norm penalty. The joint graph learning and signal inference problem is then formulated as

$$\min_{\mathbf{A},\mathbf{E},\mathbf{X}} \; \|\mathbf{E}\|_F^2 + \alpha \|\mathbf{A}\|_1 + \beta \|\mathbf{X}\|_* + \gamma g_\delta(\mathbf{X}) \tag{3a}$$

$$\text{s.t. } \mathbf{Z} = \mathbf{A}(\mathbf{Z}-\mathbf{E}) + \mathbf{X} + \mathbf{E}, \tag{3b}$$

$$A_{i,i} = 0, \; i = 1, ..., N, \tag{3c}$$

where $\mathbf{Z} = \mathbf{Y} + \mathbf{E}$ represents noisy observations and $\mathbf{E}$ is the error matrix. The term $\|\mathbf{A}\|_1 = \sum_{i,j} |A_{i,j}|$ promotes a sparse network, while $g_\delta(\mathbf{X})$ balances spatial and temporal priors

$$g_\delta(\mathbf{X}) = (1-\delta)\|\mathbf{X}\|_{1,2} + \delta \|\mathbf{X}\mathbf{D}_T\|_1, \tag{4}$$

where $\|\mathbf{X}\|_{1,2} = \sum_{i=1}^N \|\mathbf{x}_i\|_2$ and $\mathbf{x}_i^\top$ is the i-th row of $\mathbf{X}$.

B. ADMM

We develop an ADMM-based scheme to solve the optimization problem (3). We introduce auxiliary variables $\mathbf{G} = \mathbf{X}$ and $\mathbf{Y} = \mathbf{X}$, and replace $\|\mathbf{X}\|_*$ and $\|\mathbf{X}\mathbf{D}_T\|_1$ in (3) by $\|\mathbf{G}\|_*$ and $\|\mathbf{Y}\mathbf{D}_T\|_1$, respectively. The augmented Lagrangian is

$$\begin{aligned}
&\mathcal{L}_\rho(\mathbf{A},\mathbf{E},\mathbf{X},\mathbf{G},\mathbf{Y};\Phi,\Psi)\\
&= \|\mathbf{E}\|_F^2 + \alpha\|\mathbf{A}\|_1 + \beta\|\mathbf{G}\|_* + \gamma g_\delta(\mathbf{X},\mathbf{Y})\\
&+ \mathrm{Tr}\left\{\Phi^\top\left(\bar{\mathbf{A}}\mathbf{Z}_E + \mathbf{X}\right) + \Psi^\top(\mathbf{G}-\mathbf{X}) + \Xi^\top(\mathbf{Y}-\mathbf{X})\right\}\\
&+ \frac{\rho}{2}\left(\left\|\bar{\mathbf{A}}\mathbf{Z}_E + \mathbf{X}\right\|_F^2 + \|\mathbf{G}-\mathbf{X}\|_F^2 + \|\mathbf{Y}-\mathbf{X}\|_F^2\right)
\end{aligned} \tag{5}$$

where $g_\delta(\mathbf{X},\mathbf{Y}) = (1-\delta)\|\mathbf{X}\|_{1,2} + \delta\|\mathbf{Y}\mathbf{D}_T\|_1$, while Φ, Ψ, and Ξ are the Lagrangian multipliers corresponding to the constraints. Here, $\bar{\mathbf{A}} = \mathbf{A} - \mathbf{I}$ and $\mathbf{Z}_E = \mathbf{Z} - \mathbf{E}$. The ADMM alternately updates $\mathbf{A}$, $\mathbf{E}$, $\mathbf{X}$, $\mathbf{G}$, $\mathbf{Y}$, Φ, Ψ, and Ξ while fixing the other variables. In the following presentation, iteration indices are omitted for convenience.

1) Update of $\mathbf{A}$: The adjacency matrix $\mathbf{A}$ is updated by

$$\mathbf{A} = \arg\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{A}\mathbf{Z}_E - \mathbf{R}\|_F^2 + \frac{\alpha}{\rho}\|\mathbf{A}\|_1, \tag{6}$$

where $\mathbf{R} = [\mathbf{r}_1, \ldots, \mathbf{r}_N]^\top = \mathbf{Z}_E - \mathbf{X} - \frac{1}{\rho}\Phi$. Then, each row $\mathbf{a}_{\bar{i}}$ of $\mathbf{A}$ (excluding diagonal entries) is updated in parallel

$$\mathbf{a}_{\bar{i}} = \arg\min_{\mathbf{a}} \frac{1}{2}\left\|\mathbf{r}_i - \mathbf{Z}_{E,\bar{i}}^\top \mathbf{a}\right\|_2^2 + \frac{\alpha}{\rho}\|\mathbf{a}\|_1, \tag{7}$$

where $\mathbf{Z}_{E,\bar{i}}$ is $\mathbf{Z}_E$ with the i-th row removed. This is the standard Lasso model [22], which we solve using the fast iterative shrinkage-thresholding algorithm (FISTA) [23].

2) Update of $\mathbf{E}$: The error matrix $\mathbf{E}$ is updated by

$$\mathbf{E} = \arg\min_{\mathbf{E}} \|\mathbf{E}\|_F^2 + \frac{\rho}{2}\left\|\bar{\mathbf{A}}\mathbf{E} - \Theta_\rho\right\|_F^2 = \mathbf{S}_\rho^{-1}\bar{\mathbf{A}}^\top\Theta_\rho, \tag{8}$$

where $\Theta_\rho = \bar{\mathbf{A}}\mathbf{Z} + \mathbf{X} + \frac{1}{\rho}\Phi$ and $\mathbf{S}_\rho = \frac{2}{\rho}\mathbf{I} + \bar{\mathbf{A}}^\top\bar{\mathbf{A}}$.

3) Update of $\mathbf{X}$: The input matrix $\mathbf{X}$ is updated by solving

$$\mathbf{X} = \arg\min_{\mathbf{X}} \|\mathbf{X}\|_{1,2} + \frac{\lambda}{2}\left\|\bar{\mathbf{X}} - \mathbf{X}\right\|_F^2 \tag{9}$$

where $\bar{\mathbf{X}} = \frac{1}{3}\left(\mathbf{G} + \mathbf{Y} - \bar{\mathbf{A}}\mathbf{Z}_E\right) + \frac{1}{3\rho}(\Psi + \Xi - \Phi)$ and $\lambda = \frac{3\rho}{\gamma(1-\delta)}$. Similar to the update of $\mathbf{A}$, we decompose the problem into row-wise subproblems for each $\mathbf{x}_i^\top$ of $\mathbf{X}$, yielding

$$\mathbf{x}_i = \max\left(\|\bar{\mathbf{x}}_i\|_2 - \frac{1}{\lambda}, 0\right)\frac{\bar{\mathbf{x}}_i}{\|\bar{\mathbf{x}}_i\|_2}, \tag{10}$$

where $\bar{\mathbf{x}}_i^\top$ is the i-th row of $\bar{\mathbf{X}}$.

4) Update of $\mathbf{G}$: The auxiliary variable $\mathbf{G}$ is updated by

$$\mathbf{G} = \arg\min_{\mathbf{G}} \frac{\beta}{\rho}\|\mathbf{G}\|_* + \frac{1}{2}\left\|\mathbf{G} - \mathbf{X} + \frac{\Psi}{\rho}\right\|_F^2 = \mathcal{D}_{\frac{\beta}{\rho}}\left(\mathbf{X} - \frac{\Psi}{\rho}\right), \tag{11}$$

where $\mathcal{D}_\lambda(\mathbf{X}) = \mathbf{U}\mathcal{D}_\lambda(\Sigma)\mathbf{V}^\top$ is the singular-value thresholding operator [24] with $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$ as the singular value decomposition of $\mathbf{X}$, $\Sigma = \mathrm{diag}\left(\{\sigma_i\}_{i=1}^r\right)$ containing singular values σ_i, and $\mathcal{D}_\lambda(\Sigma) = \mathrm{diag}(\max(\sigma_i - \lambda, 0))$.

5) Update of $\mathbf{Y}$: Analogous to $\mathbf{X}$, the auxiliary variable $\mathbf{Y}$ can be updated row-wise

$$\mathbf{y}_i = \arg\min_{\mathbf{y}} \frac{1}{2}\|\mathbf{q}_i - \mathbf{y}\|_2^2 + \frac{\gamma\delta}{\rho}\left\|\mathbf{D}_T^\top\mathbf{y}\right\|_1, \tag{12}$$

where $\mathbf{q}_i$ is the i-th row of $\mathbf{X} - \Xi/\rho$. Since $\mathbf{D}_T$ is a difference operator, this can be reformulated as the Lasso problem

$$\mathbf{p}_i = \arg\min_{\mathbf{p}} \frac{1}{2}\|\mathbf{q}_i - \mathbf{C}\mathbf{p}\|_2^2 + \frac{\gamma\delta}{\rho}\|\mathbf{p}\|_1, \tag{13}$$

where $\mathbf{C}$ is a lower-triangular matrix of ones. We solve this using FISTA and recover $\mathbf{y}_i$ by $\mathbf{C}\mathbf{p}_i$.

6) Update of Φ, Ψ, and Ξ: The Lagrangian multipliers Φ and Ψ are updated as follows

$$\Phi \leftarrow \Phi + \rho\left(\bar{\mathbf{A}}\mathbf{Z}_E + \mathbf{X}\right), \Psi \leftarrow \Psi + \rho(\mathbf{G} - \mathbf{X}), \tag{14}$$

$$\Xi \leftarrow \Xi + \rho(\mathbf{Y} - \mathbf{X}). \tag{15}$$

The iterative procedure continues until the relative updates of $\mathbf{A}$, $\mathbf{E}$, and $\mathbf{X}$ are sufficiently small, or until the maximum number of iterations is reached. The proposed algorithm is henceforth referred to as XTLS-SEM.

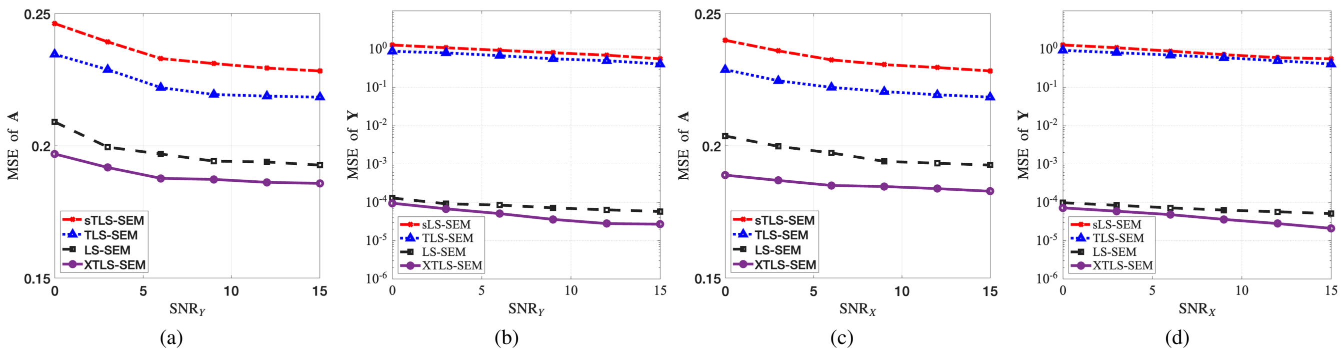


Fig. 1. Performance comparison under varying SNR.

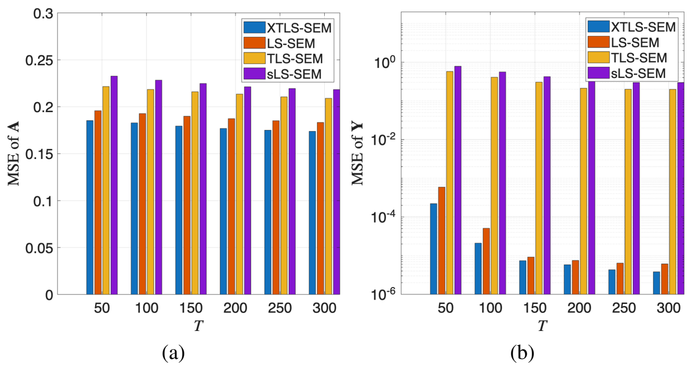


Fig. 2. Performance comparison under varying T.

III. Numerical Experiments

This section evaluates the proposed XTLS-SEM using both synthetic and real-world data. On synthetic datasets, XTLS-SEM is compared with LS-SEM [11], [12], TLS-SEM [21], and its sparse variant sTLS-SEM [21], with all hyperparameters selected via exhaustive grid search for fair comparison.

We further validate the method on a real-world epileptic EEG dataset [25], [26], where exogenous inputs are unobserved. Unlike baseline approaches whose network topology can not be estimated under missing input information, XTLS-SEM effectively recovers the network topology by jointly estimating latent signals and causal dependencies.

A. Synthetic data

We simulate a network of $N = 20$ nodes using an Erdős-Rényi (ER) graph with an edge connection probability 0.2. The system is driven by Q exogenous inputs, each affecting P randomly selected nodes. Each input spans T time steps and consists of up to three non-overlapping segments with randomly onset times and durations not exceeding $\left\lfloor \frac{T}{3} \right\rfloor$. The nonzero entries of $\mathbf{X}$ are sampled uniformly from $[-1, 1]$.

Given $\mathbf{A}$ and $\mathbf{X}$, observations are generated as $\mathbf{Z} = \mathbf{Y} + \mathbf{E}$, where $\mathbf{Y} = (\mathbf{I} - \mathbf{A})^{-1} \mathbf{X}$ and $\mathbf{E}$ represents i.i.d. Gaussian noise $\mathcal{N}\left(0, \sigma_E^2\right)$. The noise level σ_E is calibrated to achieve a target output signal-to-noise ratio (SNR), defined as $\mathrm{SNR}_Y = 10 \log_{10} \left(\frac{\|\bar{\mathbf{y}}\|_2^2}{N \sigma_E^2} \right)$ with $\bar{\mathbf{y}} = \frac{1}{T} \mathbf{Y} \cdot \mathbf{1}$. For LS-SEM, TLS-SEM, and sTLS-SEM, $\mathbf{X}$ is assumed known. To simulate realistic conditions, noisy inputs $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{R}$ are used, where $\mathbf{R}$ is scaled to achieve a target input SNR_X.

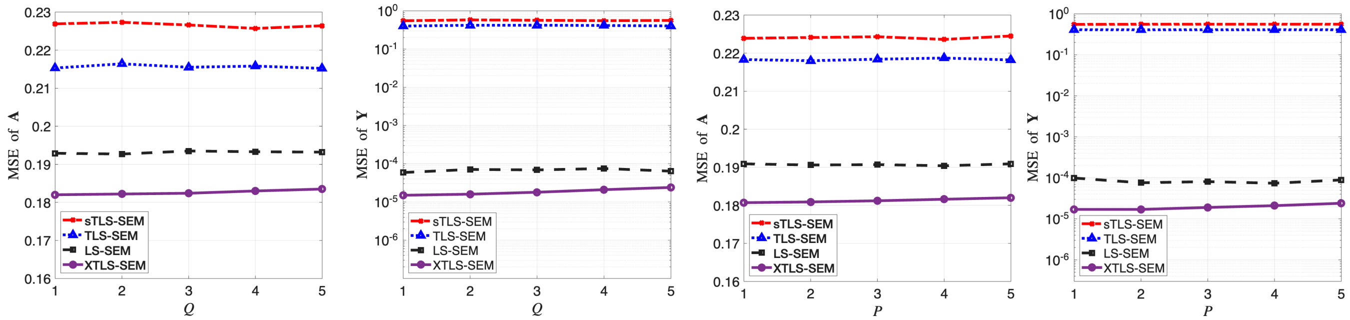
Algorithm performance is evaluated across various configurations of SNR_Y, SNR_X, Q, P, and T, with each setting repeated 100 times to ensure statistical robustness. Evaluation metrics include the mean squared error (MSE) of the adjacency matrix estimation, $\mathrm{MSE} = \frac{1}{N^2} \|\mathbf{A} - \mathbf{A}_{est}\|_F^2$, and the output reconstruction, $\mathrm{MSE} = \frac{1}{NT} \|\mathbf{Y} - \mathbf{Y}_{est}\|_F^2$, where $\mathbf{A}_{est}$ and $\mathbf{Y}_{est}$ denote the estimated adjacency and output matrices.

1) Varying input and output SNR: We evaluate the effect of SNR_Y and SNR_X while fixing the other at 15 dB ($Q = 2$, $P = 3$, and $T = 100$). Figs. 1a and 1b illustrate the adjacency matrix estimation and output reconstruction accuracy under varying observation noise levels, while Figs. 1c and 1d present the corresponding results for varying input noise. All methods benefit from increasing SNR. However, XTLS-SEM consistently achieves the lowest estimation and reconstruction errors across all noise levels, with its advantage becoming more pronounced at higher SNRs, demonstrating superior robustness to both input and output noise.

2) Varying T: We further assess algorithm's performance by varying the sample size T, while maintaining SNR_Y and SNR_X at 15 dB and keeping all other parameters constant. As illustrated in Fig. 2, increasing the number of time steps T enhances the estimation accuracy for all methods, with the most significant improvements observed in output reconstruction. Notably, XTLS-SEM consistently achieves the lowest MSE for both adjacency matrix identification and output signal estimation across the entire range of T.

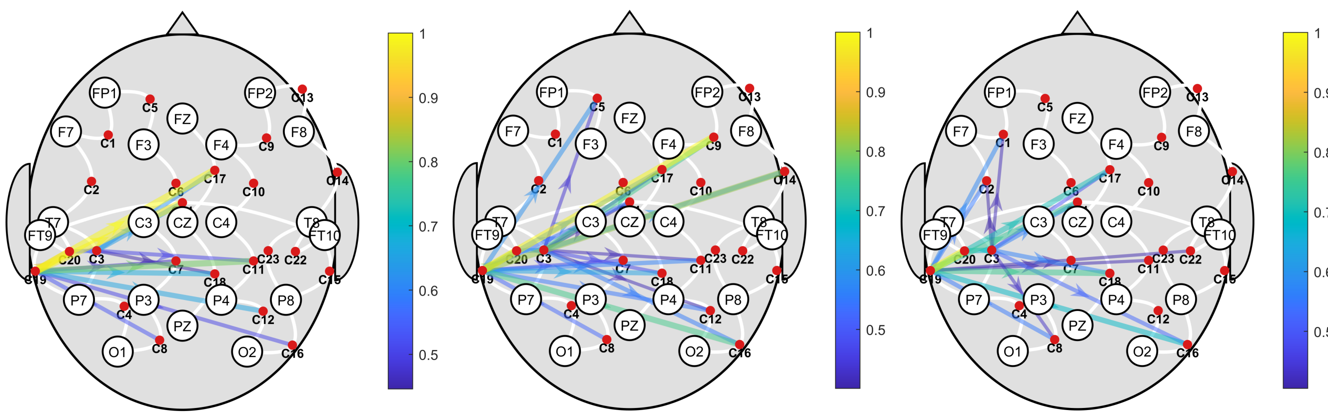
3) Varying Q and P of $\mathbf{X}$: Finally, we examine the impact of the input structure by varying the number of input channels Q and the number of nodes influenced per channel P—with $\mathrm{SNR}_X = \mathrm{SNR}_Y = 15$ dB and $T = 100$.

When varying Q with $P = 2$ (3a and 3b), most methods maintain relatively stable performance, while XTLS-SEM consistently outperforms the baselines. However, its estimation accuracy slightly decreases as Q increases. This trend is attributed to the fact that a higher Q leads to a denser exogenous input matrix $\mathbf{X}$, which gradually violates the underlying low-rank and sparsity assumptions. Similar behavior is observed



(a) (b) (c) (d)

Fig. 3. Performance comparison under varying Q or P.



(a) Pre-seizure (b) Seizure (c) Post-seizure

Fig. 4. Estimated brain network topology of the epileptic patient at different seizure stages using the proposed method.

in 3c and 3d when varying P with $Q = 3$, highlighting the sensitivity of the joint inference task to the density of $\mathbf{X}$.

B. Epileptic Dataset

The epileptic EEG data analyzed were obtained from the CHB-MIT Scalp EEG Database [25], [26], recorded from pediatric patients with intractable epilepsy. We focused on an 11-year-old female subject monitored continuously after discontinuing anti-seizure medication to characterize seizure dynamics. EEG signals ($\mathbf{Y}$) were recorded via the International $10-20$ system, resulting in 23 bipolar channels (labeled $C_1, C_2, \ldots, C_{23}$). Each channel represents the potential difference between specific electrode pairs (e.g., FP1-F7). For our analysis, all signals were downsampled to 64 Hz.

Fig. 4 illustrates the estimated brain network dependency structures at three representative stages: the pre-seizure period (Fig. 4a), the ictal (seizure) period (Fig. 4b), and the post-seizure period (Fig. 4c). As shown, the networks estimated during the pre- and post-seizure intervals exhibit relatively sparse causal connectivity, with directed edges primarily localized to specific regions. This suggests that brain electrical activity remains comparatively stable outside of the seizure window. In contrast, during the ictal stage, the estimated brain network becomes more dense and distributed. This transition indicates the rapid expansion of pathological neural interactions across multiple cortical regions. In particular, we observe a notable increase in causal dependencies originating from the left cerebral region towards channels C_9 and C_{14}. This phenomenon is consistent with the neurophysiological hallmark of epilepsy, where hyper-synchronous neural discharges propagate from a focal origin to recruit widespread cortical networks.

IV. Conclusion

In this paper, we proposed a novel SEM framework that incorporates the acting nodes and temporal durations of exogenous inputs, enabling the joint inference of directed causal dependency graphs and latent measurement signals without requiring known exogenous inputs. The resulting model provides a flexible and generalized representation suitable for complex real-world systems. An efficient ADMM-based solver was developed, with subproblems solved either in closed form or via FISTA. Numerical experiments on synthetic data demonstrate the robustness of the proposed method in jointly estimating signal structures and network topology, while experiments on epileptic EEG data reveal neurophysiologically plausible connectivity patterns, validating the effectiveness of the XTLS-SEM framework.

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